

WEEK-06

Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

```
#include <stdio.h>
```

```
#include <time.h>
```

```
void heapifyUp(int arr[], int i) {  
    while (i > 0 && arr[(i - 1) / 2] < arr[i]) {  
        int temp = arr[i];  
        arr[i] = arr[(i - 1) / 2];  
        arr[(i - 1) / 2] = temp;  
  
        i = (i - 1) / 2;  
    }  
}
```

```
void buildHeap(int arr[], int n) {  
    for (int i = 1; i < n; i++) {  
        heapifyUp(arr, i);  
    }  
}
```

```
void heapifyDown(int arr[], int n, int i) {  
    int largest = i;  
  
    while (1) {  
        int left = 2*i + 1;  
        int right = 2*i + 2;  
  
        if (left < n && arr[left] > arr[largest])  
            largest = left;  
  
        if (right < n && arr[right] > arr[largest])  
            largest = right;  
  
        if (largest != i) {  
            int temp = arr[i];  
            arr[i] = arr[largest];  
            arr[largest] = temp;  
  
            i = largest;  
        } else {  
            break;  
        }  
    }  
}
```

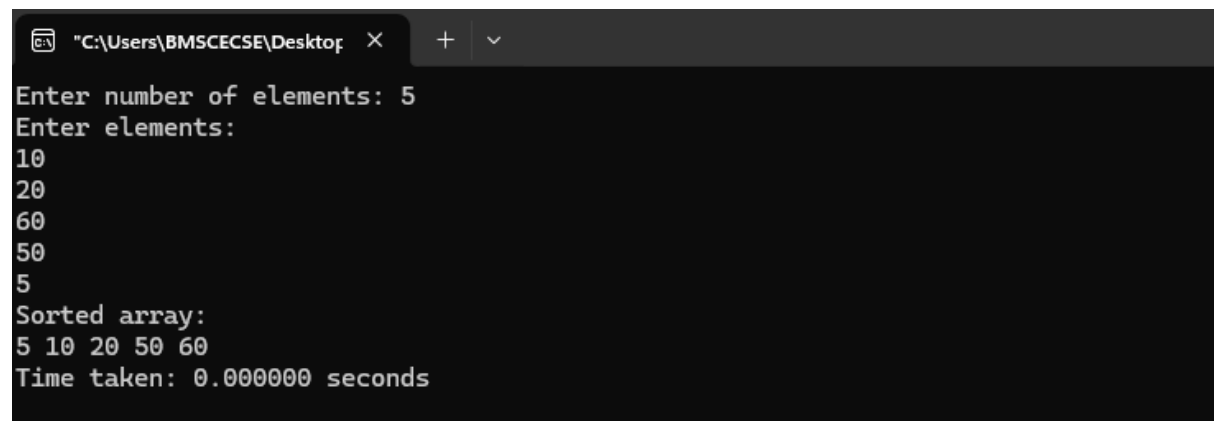
```
    }  
  }  
}
```

```
void heapSort(int arr[], int n) {  
    buildHeap(arr, n);  
    for (int i = n - 1; i > 0; i--) {  
        int temp = arr[0];  
        arr[0] = arr[i];  
        arr[i] = temp;  
  
        heapifyDown(arr, i, 0);  
    }  
}
```

```
void printArray(int arr[], int n) {  
    for (int i = 0; i < n; i++)  
        printf("%d ", arr[i]);  
}
```

```
int main() {  
    int n;  
  
    printf("Enter number of elements: ");  
    scanf("%d", &n);  
  
    int arr[n];  
  
    printf("Enter elements:\n");  
    for (int i = 0; i < n; i++)  
        scanf("%d", &arr[i]);  
  
    clock_t start = clock();  
  
    heapSort(arr, n);  
  
    clock_t end = clock();  
  
    printf("Sorted array:\n");  
    printArray(arr, n);  
  
    double time_taken = (double)(end - start) / CLOCKS_PER_SEC;  
    printf("\nTime taken: %f seconds\n", time_taken);  
  
    return 0;  
}
```

Output:



```
"C:\Users\BMSCECSE\Desktop" X + v
Enter number of elements: 5
Enter elements:
10
20
60
50
5
Sorted array:
5 10 20 50 60
Time taken: 0.000000 seconds
```

The screenshot shows a Windows command prompt window with a single tab titled '"C:\Users\BMSCECSE\Desktop" X'. The window has a dark background and displays the following text: 'Enter number of elements: 5', 'Enter elements:', followed by five lines of input values (10, 20, 60, 50, 5), then 'Sorted array:', followed by the output '5 10 20 50 60', and finally 'Time taken: 0.000000 seconds'.